

Bobby Cheema

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EDUCATION

University of California, Berkeley

Berkeley, CA

B.S. Electrical Engineering and Computer Science (EECS)

May 2029

- **Relevant Coursework:** Structure and Interpretation of Computer Programs, Data Structures, Linear Algebra & Differential Equations, Computer Architecture, Introduction to Signals, Dynamics, and Information

EXPERIENCE

Software Engineering Intern

June 2026 – August 2026

NASA Ames Research Center | Multi-Mission Operations Center (MMOC)

Mountain View, CA

- Built a centralized infrastructure monitoring dashboard aggregating real-time health and performance data across 50+ infrastructure hosts, providing a unified view of system health, availability, and operational status.
- Designed data aggregation and alert-processing pipelines to consolidate 300+ service health checks into actionable infrastructure metrics, accelerating detection of outages and performance anomalies.
- Developed interactive visualizations for uptime, latency, service availability, and historical alert trends, reducing manual monitoring and improving infrastructure observability.

Web Developer

Feb 2026 – May 2026

Moffitt Status | ASUC, UC Berkeley

Berkeley, CA

- Developed a production full-stack application serving 1,000+ UC Berkeley students, enabling near real-time querying of library facility availability data
- Designed backend pipelines to convert 2D library layout plans into 3D spatial models, experimenting with data representations and export formats for navigation features
- Optimized 3D model generation workflows, reducing transformation time by 30% and improving pipeline efficiency for frontend visualization

Software Engineering Intern

September 2025 – December 2025

Second Foundation AI

San Francisco, CA

- Built an agentic AI dashboard orchestrating autonomous GPT-4 agents for multi-step reasoning, generating 100+ automated business intelligence insights per session
- Developed multi-agent task planning and dynamic prompt engineering pipelines using Python, FastAPI, and Plotly, reducing end-to-end query latency by 35% and supporting real-time analytics
- Implemented evaluation pipelines to measure agent performance, achieving 85% task success rate while tracking latency, error rates, and decision consistency across runs

PROJECTS

Nexus | *Python, Flask, PostgreSQL, D3.js, GPT-4/Claude*

January 2026

- Collaborated in a backend engineering team to migrate 7,800+ company records from fragmented JSON files to a normalized PostgreSQL schema with graph-modeled relationships
- Built RESTful APIs and an interactive D3.js force-directed graph to enable real-time exploration of competitors, suppliers, and subsidiaries
- Integrated GPT-4/Claude APIs and ETL pipelines to auto-generate company relationships, improving data completeness and reducing manual entry by 60%

Smart Sort | *Python, PyTorch, JavaScript (Canvas API)*

June 2025

- Trained a PyTorch neural network to learn optimized sorting heuristics on synthetically generated data arrays.
- Evaluated learned models against traditional algorithms by visualizing runtime and stability metrics using the JavaScript Canvas API.
- Optimized training efficiency by experimenting with different loss functions and data distributions, improving model stability across test cases.

SKILLS

Languages: Python, C/C++, Java, TypeScript/JavaScript, HTML/CSS, SQL

Frameworks: PyTorch, NumPy, Pandas, FastAPI, Flask, React, Node.js, Express, Plotly, JUnit

Developer Tools: Git, GitHub, Docker, VS Code, Linux, Bash